INCH-POUND

MIL-M-28787/86B
27 November 1991
SUPERSEDING
MIL-M-28787/86A(EC)
10 April 1979

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RELAY, POWER, KEY CODE ERN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic power relay module.

1.2 Module description. The following electrical parameters reflect the 0 to 60°C end-of-life limits in table I.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

Name	Relay, power
Family	Miscellaneous
Key code	ERN
Size	1F
Weight	284 grams maximum
Failure rate	0.253 failures/10^6 hours maximum plus 0.100 failures/10^6 contact operations
Environment	Class I as specified in MIL-STD-1389
Module input voltage and current for coil operation	10 Vdc ±5 percent, 223 mA ±25 percent 25 Vdc ±10 percent, 223 mA ±25 percent 32 Vdc ±10 percent, 223 mA ±25 percent

Maximum Voltage

Pins +	Pins -	Volts DC
1A1	1A31	12.0
1A1	1B21	32.5
1A1	1B14	32.5
1A1	1B34	32.5

Pickup voltage (pins 1A1 and 1A31)	3.0 Vdc minimum, 8.7 Vdc maximum
Dropout voltage (pins 1A1 and 1A31)	0.5 Vdc minimum, 2.9 Vdc maximum
Operate time (including contact bounce)	3 mS minimum, 20 mS maximum
Release time (including contact bounce)	2 mS minimum, 15 mS maximum
Slow release time (including contact bounce)	6 mS minimum, 25 mS maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

Contact resistance	0.018 Ohm maximum
Contact voltage	33.0 Vdc maximum (at 20 A maximum) 35.0 Vdc maximum (at 2 A maximum) 115.0 VAC, 5.1 KHz (at 1 A maximum
Contact current	50.0 amperes maximum for 1.0 mS maximum (resistive load, 1 percent duty cycle)
Carrying current	
Single contact usage	35 A maximum
Dual contact usage	25 A per contact maximum
Maximum power dissipation	30 W
Cooling	Cooling shall be provided to insure fin does not exceed 60° C

1.3 Module function. The module contains a three pole, double throw, break before make relay with high current and voltage switching capability. Programmable resistors permit relay power dissipation reduction at operating or holding voltages up to 35.5 Vdc.

1.3.1 Input voltage. The voltage input to the module for relay operation shall be 10 Vdc ±5 percent, 25 Vdc ±5 percent, or 32 Vdc ±5 percent.

1.3.2 Transient suppression. Inductive transients from the relay coil may be suppressed by programming the internal suppression circuit across the coil.

1.3.3 Maximum contact rating. The relay module contact rating shall be 20 amperes at 33 Vdc +0 percent, -5 percent resistive load, or 2.0 amperes ± 20 percent at 35 Vdc +0 percent, -10 percent resistive load.

1.3.3.1 Module pins. All module pins assigned to each relay contact must be connected in shunt for all contact current conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.6).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.253 failures per million operating hours and 0.100 failures per million contact operations when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be ERN.

3.6.2 Size. The module size shall be 1F.

3.6.3 Weight. The module shall have a maximum weight of 284 grams.

3.6.4 Contact pins. The module shall have 80 contact pins positioned A1 through A40 and B1 through B40.

3.6.5 Fin position. The module shall have an insertion and extraction fin on increments A and B, and heat dissipation fins on increments A through F.

3.6.6 Depth. The module shall have a depth of 0.210 $\pm$ 0.005 inches (5.334 $\pm$ 0.127 mm) in module increments C, D, E, and F.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/86-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.7 Circuit pin configuration. The functional block diagram and pin assignments shall conform to figure 1.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

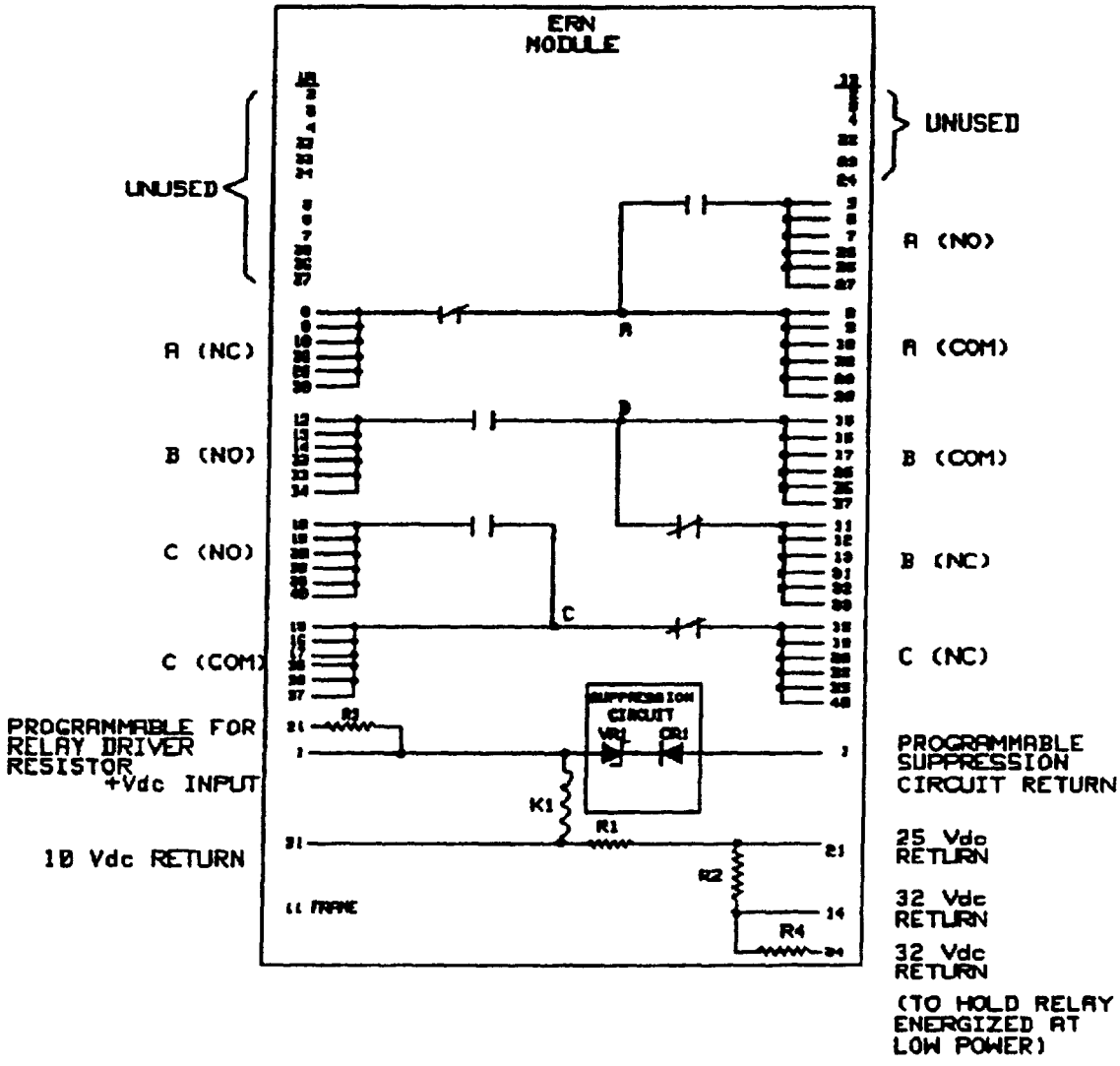
3.10 Preconditioning. Prior to any testing, each module shall be preconditioned in accordance with 4.4. When tested in accordance with 4.4, a contact resistance greater than 1.0 Ohm for a closed contact during any cycle shall be cause to reject that module (or relay). Rejected modules or relays shall not be shipped.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.



NOTES:

1. R3 = 825 Ohm ± 1 percent, 1/2 W.
2. R1 = 65.5 Ohm ± 11.5 percent, 7 W minimum.
3. R2, R4 = 47.0 Ohm ± 5 percent, 3 W.

FIGURE 1. ERN module function diagram power relay.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25 ± 5°C Ambient Temperature				Operating temperature 0 to 55°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A20	Insulation resistance	II	S	100.0	INF	100.0	INF	100.0	INF	100.0	INF	MOhm
B	Relay coil resistance	III	X	42.7	52 8	42.7	52.8	33.8	58.8	33.8	58.8	Ohm
C1	Resistance R3	IV	X	817.0	833.0	782.0	869.0	800.0	850.0	785.0	885.0	Ohm
C2	Resistance R1	IV	X	58.0	71.0	58.0	74.0	58.0	74.0	57.4	75.0	Ohm
C3,C4	Resistance R2,R4	IV	X	44.6	48.4	43.0	51.0	43.0	51.0	42.0	52.0	Ohm
D1-D8	Frame continuity	V	S	0.0	1 0	0.0	1.0	0.0	1 0	0.0	1.0	Ohm
E1-E45	Pin jumper continuity	VI	X	0.0	00.0	0.0	00.0	0.0	00.0	0.0	00.0	mV
E45	Pin jumper continuity balance	VI	X	0.0	30.0	0.0	30.0	0.0	30.0	0.0	30.0	mV
F1-F12	Isolation	VII	X	10.0	INF	1.0	INF	10.0	INF	1.0	INF	MOhm
G1	Suppression circuit reverse leakage	VIII	S	0.0	10.0	0.0	10.0	0.0	10.0	0.0	10.0	uA
G2	Suppression circuit breakdown voltage	VIII	X	30.0	33.0	30.0	33.0	30.0	33.0	30.0	33.0	V
H1-H6	Contact voltage drop (2 to 20 A)	IX	X	0.0	10.0	0.0	10.0	0.0	10.0	0.0	10.0	mV/A
J1	Relay pickup voltage	X	S	2.0	7 2	2.0	8.2	2.0	6 4	2.0	8.7	V
J2	Relay pickup voltage	X	X	12.0	29 0	12.0	28.0	12.0	29 0	12.0	28.0	V
J3	Relay pickup voltage	X	X	10.0	22.0	10.0	22.0	10.0	22.0	10.0	22.0	V
J4	Relay pickup voltage	X	S	9.0	20.0	9.0	20.0	9.0	20.0	9.0	20.0	V
K1	Relay dropout voltage	XI	S	0.5	2.9	0.5	2.9	0.5	2.8	0.5	2.9	V
K2	Relay dropout voltage	XI	S	0.5	11.0	0.5	11.0	0.5	11.0	0.5	11.0	V
K3	Relay dropout voltage	XI	S	0.5	9 0	0.5	9.0	0.5	9.0	0.5	9.0	V
K4	Relay dropout voltage	XI	S	0.5	0.0	0.5	0.0	0.5	0 0	0.5	0.0	V
L1-L9	Relay operate time (including contact bounce)	XII	X	3.0	20.0	3.0	20.0	3.0	20.0	3.0	20.0	mS

-See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25 ± 3°C Ambient Temperature				Operating temperature 0 to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M3	Relay release time (including contact bounce)	XIII	X	2.0	15.0	2.0	15.0	2.0	15.0	2.0	15.0	ms
N1-N3	Relay slow release time (including contact bounce)	XIV	X	6.0	25.0	6.0	25.0	6.0	25.0	6.0	25.0	ms
P1-P5	Contact carry current	XV	Q	25 a	-	25.0	-	25.0	-	25.0	-	a
Q1-Q3	Relay break before make time	XVI	Q	0.1	5.0	0.1	5.0	0.1	5.0	0.1	5.0	ms

-See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Preconditioning. A 5,000 cycle run-in test shall be conducted under the following conditions (the run-in test may be performed on the relay prior to installation in the module):

(a) Open circuit contact voltage shall be 6 volts (ac or dc) maximum.

(b) Contact current shall be 10 mA (ac or dc) maximum.

(c) Cycling rate shall be 10 Hz maximum and the on-off periods shall be equal within 10 percent.

(d) Coil voltage at pins 1A1 (+) and 1A31 (-) with pin 1B1 tied to 1B21 shall be 10 Vdc ± 5 percent.

(e) A contact resistance greater than 1.0 Ohm for a closed contact during any cycle shall be cause to reject that relay. Rejected relays shall not be shipped.

4.5 Test procedure. The module shall be tested as specified in table I.

4.5.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules (SEM) Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.5.2 Order of testing. Test conditions A through Q shall be performed in any order with the following exceptions: If test conditions A or F are included in a test sequence, the order of testing shall be as follows: A, F, D, E, B and C where applicable, followed by the remaining tests in the sequence.

4.6 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

CAUTION. If no external cooling is used, power shall not be applied to the module in an ambient of 25°C for a period of time in excess of 10 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

4.6.1 Qualification sample size. Sample size shall be in accordance with MIL-M-28787. Assignment of modules for overload and overcurrent tests shall be in accordance with table XVII.

4.6.2 Shock test. Shock testing shall be in accordance with MIL-M-28787. There shall be no closing of open contacts or opening of closed contacts for a time greater than 10 microseconds. To test the normally open contacts apply 25 Vdc ± 5 percent across pins 1A1 (+) and 1B21 (-). Following the shock test, the electrical tests in table I marked with an (X) or (S) in the test type column shall be performed using initial limits at 25° ± 5°C ambient.

4.6.3 Vibration test. Vibration testing shall be in accordance with MIL-M-28787. Apply 25 Vdc ± 5 percent across pins 1A1 (+) and 1B21 (-) to test the normally open contacts. During vibration testing, closed contacts shall have a contact resistance of 0.10 Ohm maximum. The relay shall be energized during one-half of each test cycle with a coil voltage of 25 Vdc ± 5 percent across pins 1A1 (+) and 1B21 (-) with 1B1 connected to pin 1B21. Following the vibration test, the electrical tests in table I marked with an (X) or (S) in the test type column shall be performed using the initial limits at 25° ± 5°C ambient.

4.6.4 Over-current tests. The module shall be energized with 12.5 ± 0.1 V coil voltage from pin 1A1 (+) to 1A31 (-) at 60 ± 5°C (fin temperature) and each relay contact shall carry a resistive load of 20 amperes at 33 Vdc +0, -5 percent for a period of 72 hours. Pins 1B1 and 1B21 shall be jumpered during all testing. Upon removal of the coil voltage after the 72-hour period, the release time shall be as specified in the 0 - 60°C EOL limits in table I, for test conditions M1 through M3.

4.6.5 Over-load test. Switch each contact 10 times with 25 +0, -0.5 amperes resistive load at 33 Vdc +0, -5 percent. Pins 1B1 and 1B21 shall be jumpered during all testing. The fin temperature shall be 60 ± 5°C. Following the test, perform test conditions H1 through H6 in table I. Values shall be as specified for 25 ± 5°C ambient EOL requirements.

4.6.6 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Q shall be performed in the order specified in 4.5.2 using the initial limits at 25 ± 5°C. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, C, and G through N shall be performed in the order specified in 4.5.2 using the initial limits for the applicable operating temperature. At all other times during the

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

operating temperature tests, the modules shall be operated as shown on figure 16.

4.6.7 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated in the test circuit of life test 1 in accordance with 4.6.8.5.

4.6.8 Life tests.

4.6.8.1 Life-test coil voltage. The coil voltage for all life tests shall be 25 Vdc across pins 1A1 (+) and 1B21 (-) with pin 1B1 connected to pin 1A31 (see figure 16).

4.6.8.2 Life-test fin temperature. The fin temperature shall be 60 ± 5°C for all life tests.

4.6.8.3 Life-test contact measurement. Test equipment shall monitor the closing of each contact being tested during the life tests.

4.6.8.4 Life-test failure. Any out of limit post life measurement shall be considered a module failure.

4.6.8.5 Life-test 1. All contacts shall be subjected to 10,000 cycles at a rate of 10 plus or minus 2 cycles per minute while switching 20 amperes resistive load at 33 Vdc +0, -5 percent, using the test circuit shown on figure 16.

4.6.8.5.1 Life-test 1 failure. A module failure shall be defined as a maximum of four operations in which the dynamic resistance of a closed contact exceeds 0.100 Ohm. Following the test of 4.6.8.5, with the temperature maintained at 60°C, electrical tests A, B, C, E, and G through N, shall be performed in the order specified in 4.5.2 using the end-of-life operating temperature requirements in table I. The module temperature shall then be reduced to 25°C in accordance with MIL-M-28787. Following the return of the module temperature to 25°C, electrical tests A through E and G through N shall be performed in the order specified in 4.5.2 using the 25 ± 5°C end-of-life requirements in table I.

4.6.8.6 Life-test 2. All contacts shall be subjected to 50,000 cycles at a rate of 20 plus or minus 2 cycles per minute while switching a 2.0 amperes ± 5 percent resistive load at 33 Vdc +0, -5 percent, using the test circuit shown on figure 16.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

4.6.8.6.1 Life-test 2 failure. A module failure shall be defined as a maximum of four operations in which the dynamic resistance of a closed contact exceeds 3.0 Ohms. Following the test of 4.6.8.6, with the temperature maintained at 60°C, electrical tests A, B, C, E, and G through N shall be performed in the order specified in 4.5.2 using the end-of-life operating temperature requirements in table I. The module temperature shall then be reduced to 25°C in accordance with MIL-M-28787. Following the return of the module temperature to 25°C, electrical tests A through E and G through N shall be performed in the order specified in 4.5.2 using the 25 ± 5°C end-of-life requirements in table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Cross reference of classification. Part number M28787/86-1 supersedes part number M28787/86. The two parts are interchangeable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0067)

Agent: NW

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE II. Insulation resistance test.

Test condition	Parameter	Connect A to pin	Connect B to A connector pins	Connect B to B connector pins	Initial limits 25 ± 5°C Ambient Min	Initial limits 25 ± 5°C Ambient Max	Unit	Test figure	Note
A1	Insulation resistance	1A2	3,4,22,23,24,5,6,7,25,26,27,8,12,18,15,21,11,31	2,3,4,22,23,24,5,15	100	INF	MOhm	2	1
A2		1A3	4,22,23,24,5,6,7,25,26,27,8,12,18,15,21,11,31						
A3		1A4	22,23,24,5,6,7,25,26,27,8,12,18,15,21,11,31						
A4		1A22	23,24,5,6,7,25,26,27,8,12,18,15,21,11,31						
A5		1A23	24,5,6,7,25,26,27,8,12,18,15,21,11,31						
A6		1A24	5,6,7,25,26,27,8,12,18,15,21,11,31						
A7		1A5	6,7,25,26,27,8,12,18,15,21,11,31						
A8		1A6	7,25,26,27,8,12,18,15,21,11,31						
A9		1A7	25,26,27,8,12,18,15,21,11,31						
A10	Insulation resistance	1A26	26,27,8,12,18,15,21,11,31	2,3,4,22,23,24,5,15	100	INF	MOhm	2	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE II. Insulation resistance test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to A connector pins	Connect B to B connector pins	Initial limits 25 ± 5°C Ambient Min	Initial limits 25 ± 5°C Ambient Max	Unit	Test figure	Note
A11	Insulation resistance	1A26	27,8,12,18,19,21,11,31	2,3,4,22,23,24,5,15	100	INF	MOhm	2	1
A12		1A27	8,12,18,19,21,11,31						
A13		1A8	12,18,19,21,11,31						
A14		1A12	18,19,21,11,31						
A15		1A18	19,21,11,31						
A16		1A19	21,11,31						
A17		1A21	11						
A18		1A11	31	2,3,4,22,23,24,5,15					
A19		1B2	31	3,4,22,23,24,5,15					
A20		1B3	31	4,22,23,24,5,15					
A21		1B4	31	22,23,24,5,15					
A22		1B22	31	23,24,5,15					
A23		1B23	31	24,5,15					
A24		1B24	31	5,15					
A25		1B5	31	15					
A26		1A6	31	8					1,2
A27		1B15	31	11					
A28	Insulation resistance	1A15	31	18	100	INF	MOhm	2	1,2

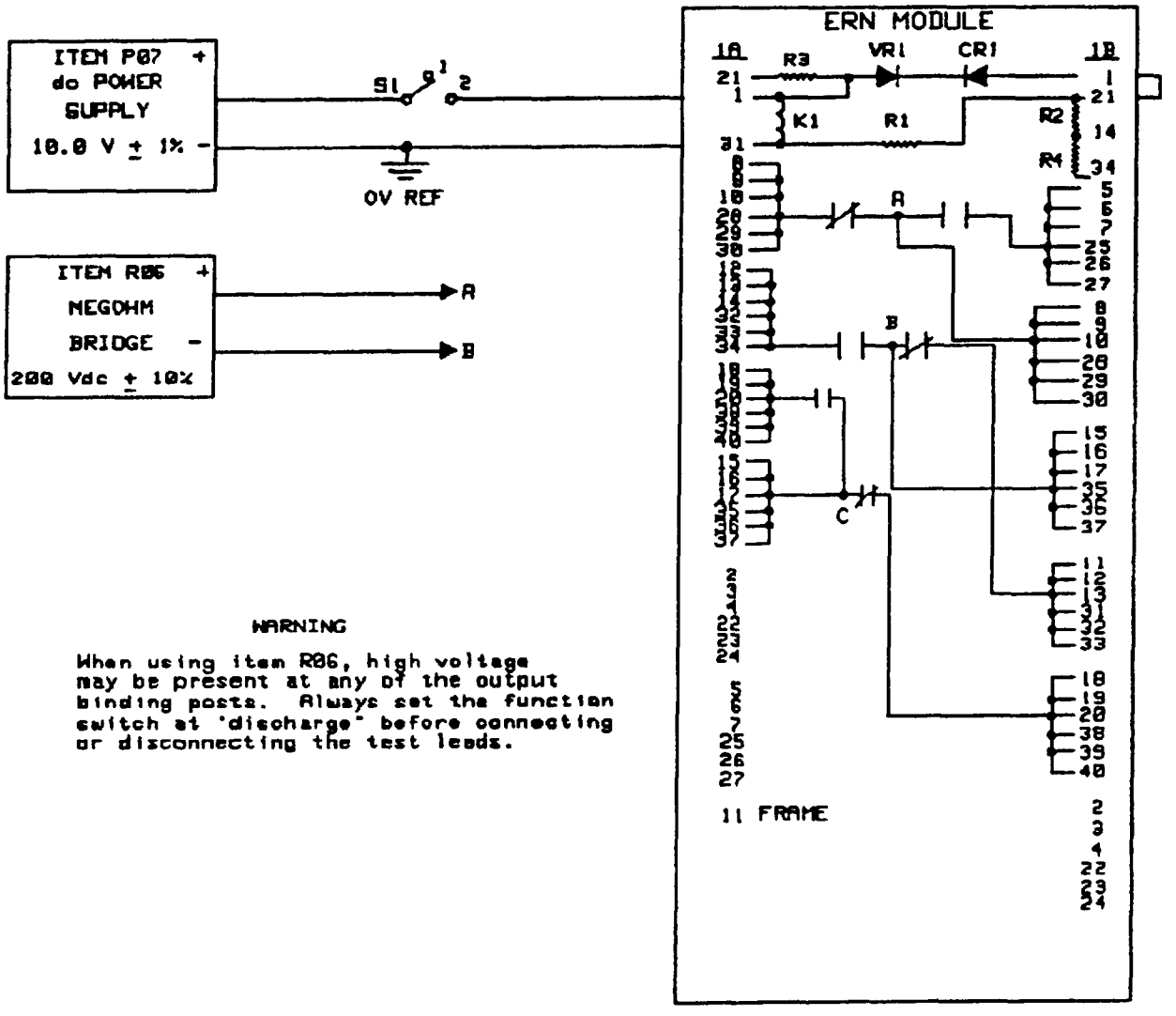
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

NOTES:

1. Pins 1B1 and 1B21 shall be jumpered throughout all testing.

2. To test the normally closed contacts, test conditions A26, A27, and A28, place S1 in position 2 (see figure 2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.



NOTE:

1. S1 = SPDT switch.

FIGURE 2. Insulation resistance test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE III. Relay coil resistance test.

Test condition	Parameter	Connect ohmmeter Item R07		Initial limits 25 ± 5°C Ambient		Unit
		+ Terminal to pin	- Terminal to pin	Min	Max	
B	Coil resistance	1A1	1A2	42.7	52.8	Ohm

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE IV. Measurement of resistors test.

Test condition	Parameter	Connect ohmmeter Item R07		Initial limits 25 ± 5°C Ambient		Unit
		+ Terminal to pin	- Terminal to pin	Min	Max	
C1	Resistance R3	1A1	1A21	817.0	833.0	Ohm
C2	Resistance R1	1A31	1B21	58.0	73.0	Ohm
C3	Resistance R2	1B21	1B14	44.6	49.4	Ohm
C4	Resistance R4	1B14	1B34	44.6	49.4	Ohm

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE V. Frame continuity test.

Test condition	Parameter	Connect ohmmeter Item R07		Initial limits 25 ± 5°C Ambient		Unit
		+ Terminal	- Terminal to module fin	Min	Max	
D1	Frame continuity	1A11	A	0.0	1.0	Ohm
D2			B			
D3			C			
D4			D			
D5			E			
D6	Frame continuity	1A11	F	0.0	1.0	Ohm

NOTE:

1. Penetrate the anodized finish on the module frame with a sharply pointed probe. Connect the negative lead of the ohmmeter to the probe.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE VI. Pin jumper continuity test.

Test condition	Parameter	Connect		Initial limits 25 ± 5°C Ambient		Units	Test figure
		A to pin	B to pin	Min	Max		
E1	Pin jumper continuity	1B6	1B5	0.0	85.0	mV	3
E2		1B7	1B5				
E3		1B25	1B5				
E4		1B26	1B5				
E5		1B27	1B5				
E6		1B9	1B8				
E7		1B10	1B8				
E8		1B28	1B8				
E9		1B29	1B8				
E10		1B30	1B8				
E11		1A9	1A8				
E12		1A10	1A8				
E13		1A28	1A8				
E14		1A29	1A8				
E15		1A30	1A8				
E16		1A13	1A12				
E17		1A14	1A12				
E18		1A32	1A12	0.0	85.0	mV	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE VI. Pin jumper continuity test - Continued.

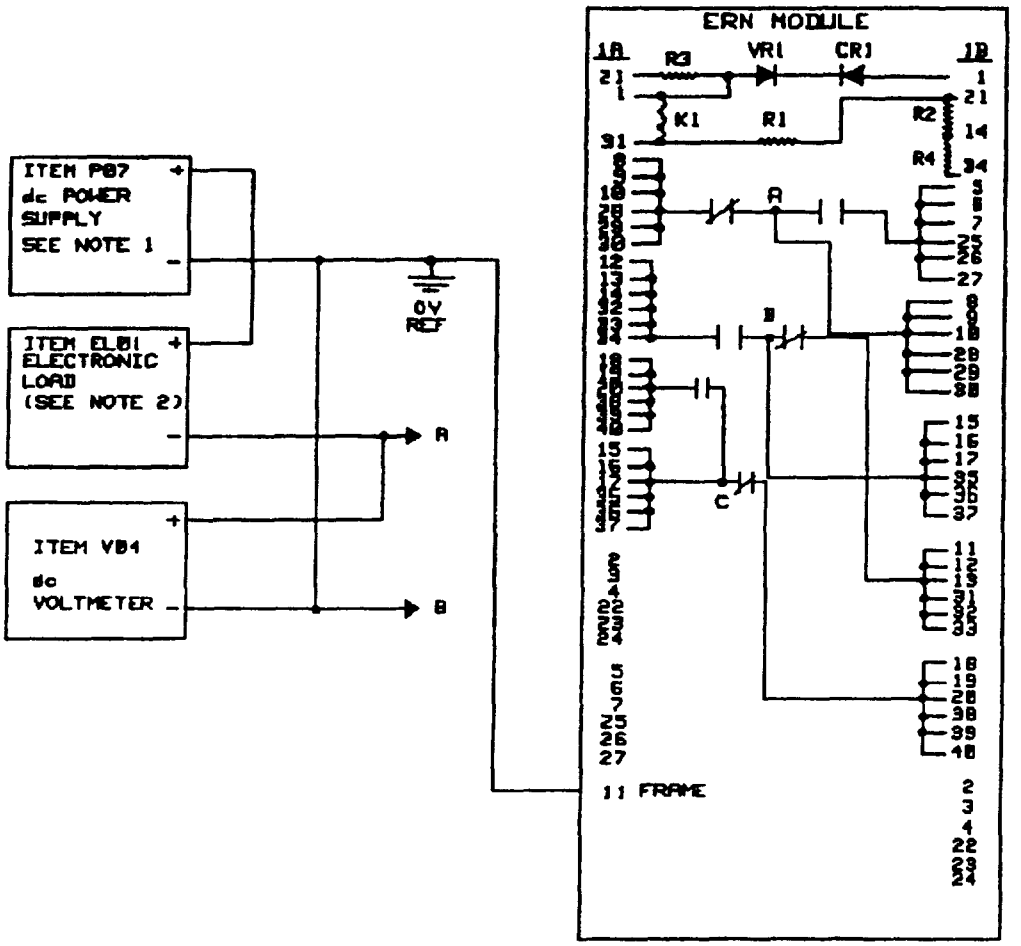
Test condition	Parameter	Connect		Initial limits 25 ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E19	Pin jumper continuity	1A33	1A12	0.0	85.0	mV	3
E20		1A34	1A12				
E21		1B16	1B15				
E22		1B17	1B15				
E23		1B35	1B15				
E24		1B36	1B15				
E25		1B37	1B15				
E26		1B12	1B11				
E27		1B13	1B11				
E28		1B31	1B11				
E29		1B32	1B11				
E30		1B33	1B11				
E31		1A19	1A10				
E32		1A20	1A18				
E33		1A38	1A18				
E34		1A39	1A18				
E35		1A40	1A18				
E36	Pin jumper continuity	1A16	1A15	0.0	85.0	mV	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE VI. Pin jumper continuity test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25 ± 3°C Ambient Min	Initial limits 25 ± 3°C Ambient Max	Unit	Test figure
E37	Pin jumper continuity	1A17	1A15	0.0	85.0	mV	8
E38		1A35	1A13				
E39		1A36	1A15				
E40		1A37	1A15				
E41		1B19	1B18				
E42		1B20	1B18				
E43		1B38	1B18				
E44		1B39	1B18				
E45	Pin jumper continuity	1B40	1B18	0.0	85.0	mV	8
E46	Pin jumper continuity balance	See note 4 figure 3	See note 4 figure 3	0.0	35.0	mV	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.



NOTES:

1. Adjust item P07 for 6.0 ± 1.0 V.

2. Connect points A and B as specified in table VI. Set item EL01 to the 0-10A mode and adjust point A to 3.0 ± 0.1A. The reading on item V04 shall be as specified in table VI.

3. Measure voltage drop (item V04) on module pins within 0.5 inch (12.7 mm) of each module pin under test.

4. Record the highest and lowest reading taken for test conditions E1-E45. The difference shall meet the limits specified in table VI, test condition E46.

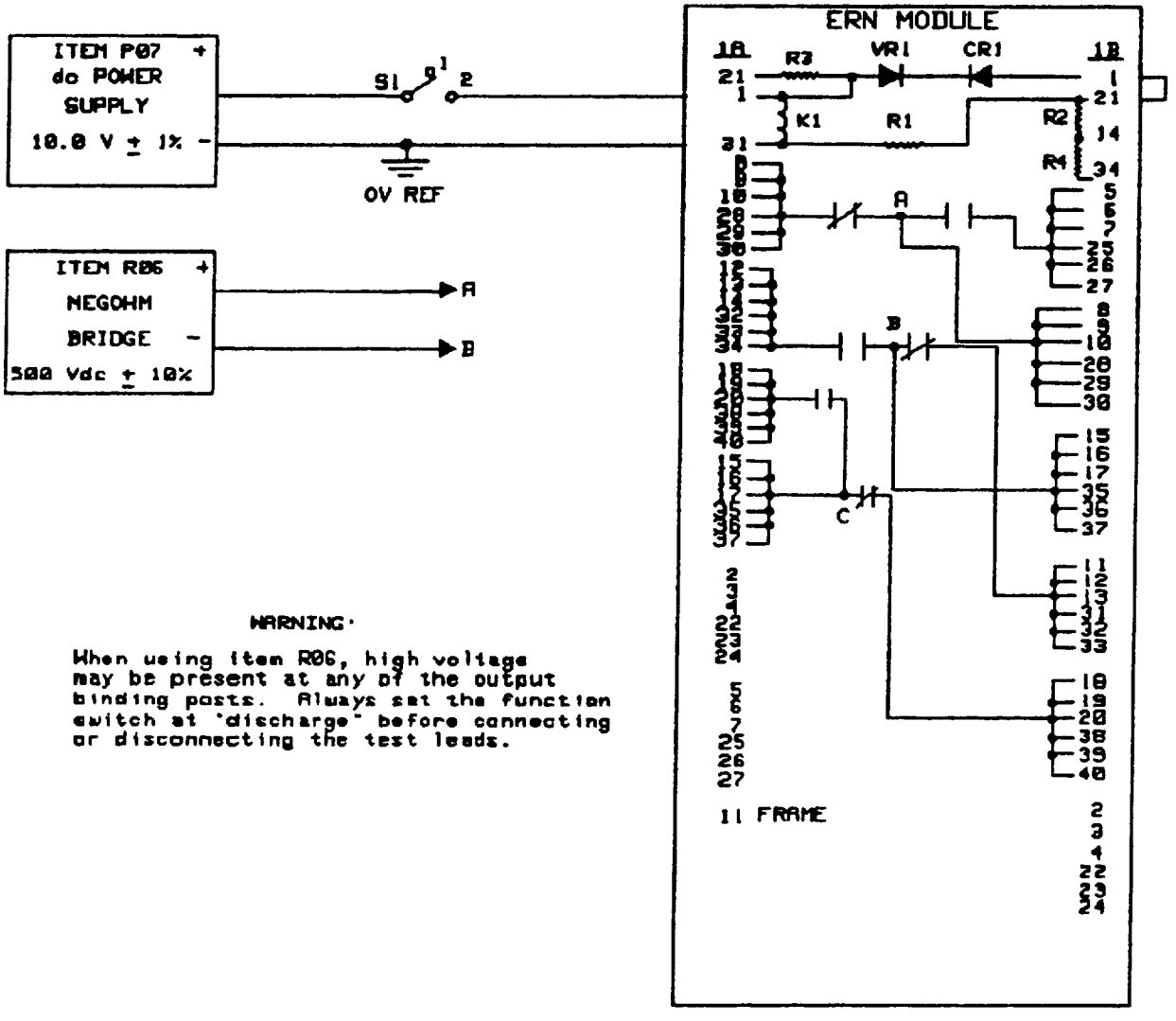
FIGURE 3. Pin jumper continuity test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE VII. Isolation test.

Test condition	Parameter	SJ position	Connect		Initial limits 25 ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F1	Isolation	1	1B5	1B8	10.0	INF	MOhm	4
F2		1	1A12	1B15				
F3		1	1A15	1A18				
F4		2	1A8	1B8				
F5		2	1B11	1B15				
F6		2	1A15	1B18				
F7		1	1B5, 1B8	1A12, 1B15				
F8			1A15, 1A18	1A12, 1B15				
F9			1A15, 1A18	1B5, 1B8				
F10			1B5, 1B8, 1B19, 1B1, 1A1, 1A15, 1A18, 1A12	1A2, 1A3, 1A4, 1A22, 1A23, 1A24, 1A5, 1A6, 1A7, 1A25, 1A26, 1A27, 1B2, 1B3, 1B4, 1B23, 1B24, 1B22				
F11			1A11	1A2, 1A3, 1A4, 1A22, 1A23, 1A24, 1A5, 1A6, 1A7, 1A25, 1A26, 1A27 1B2, 1B3, 1B4, 1B22, 1B23, 1B24, 1B5, 1B8, 1B15, 1B1, 1A1, 1A15, 1A18, 1A12				
F12	Isolation	1	1A1, 1B1, 1A31, 1B21, 1B34	1B5, 1B8, 1B15, 1A15, 1A18, 1A12	10.0	INF	MOhm	4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.



NOTE:

1. S1 = SPDT switch.

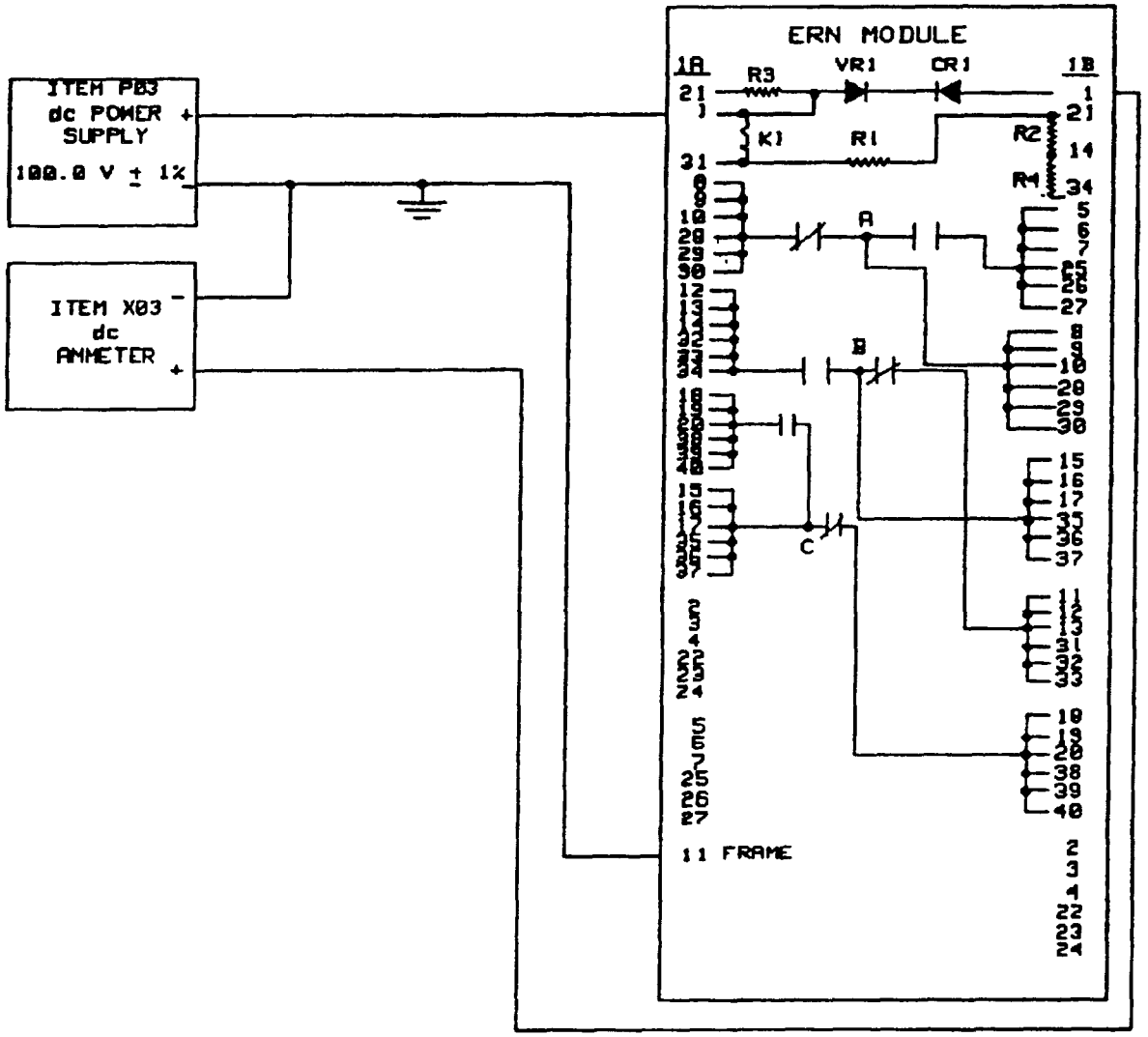
FIGURE 4. Isolation test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE VIII. Suppression circuit characteristic test.

Test condition	Parameter	Initial limits 25 ± 5°C Ambient		Unit	Test figure
		Min	Max		
G1	Reverse leakage	0.0	10.0	uA	5
G2	Breakdown voltage	30.0	39.0	V	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

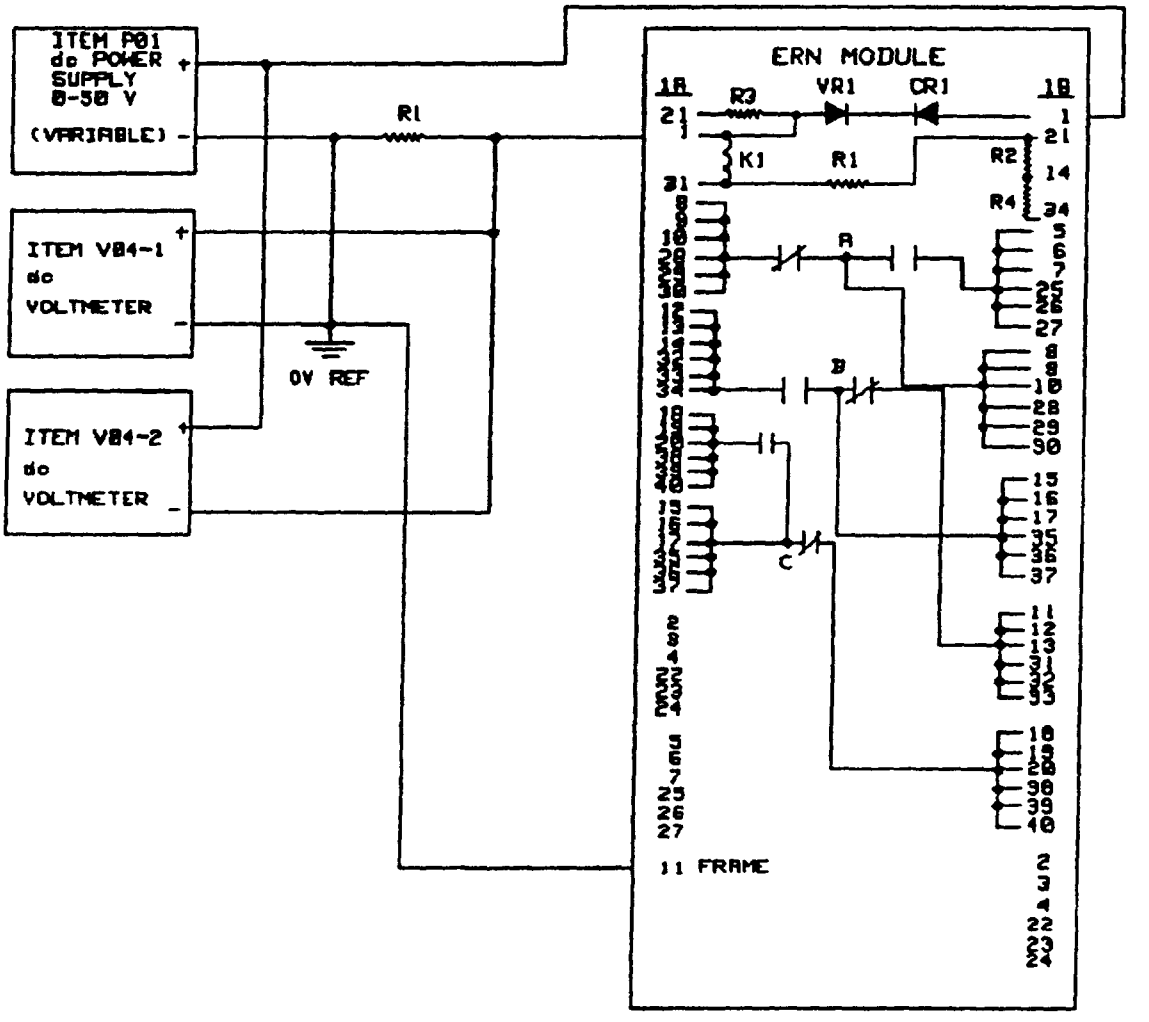


NOTE:

1. Read leakage current on X03.

FIGURE 5. Suppression circuit characteristics, reverse leakage test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.



NOTES:

1. R1 = 1 KOhm ± 5 percent, 1/4 W.

2. Increase item P01 voltage from 0 Vdc until the voltage on item V04-1 reads 7.5 Vdc ± 1 percent. Do not increase item P01 voltage over 50 Vdc. The voltage read on item V04-2 shall satisfy the requirements for test condition G2 of table VIII.

3. Since the voltmeters are not used simultaneously, only one item V04 is required.

FIGURE 6. Suppression circuit characteristics, breakdown voltage test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE IX. Contact voltage drop test.

Test condition	Parameter	Connect				Initial limits 25 ± 5°C Ambient		Unit (see note)	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Contact voltage drop A(NC)	B8	A8	B28	A28	0.0	16.0	mV/A	7
H2	B(NC)	B15	B11	B35	B31				
H3	C(NC)	A15	B18	A35	B38				
H4	A(NO)	B5	B8	B25	B28				
H5	B(NO)	A12	B15	A32	B35				
H6	Contact voltage drop C(NO)	A18	A15	A38	A35	0.0	16.0	mV/A	7

NOTE.

1. mV/A, where A = V_{R1} /1.4 Ohm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

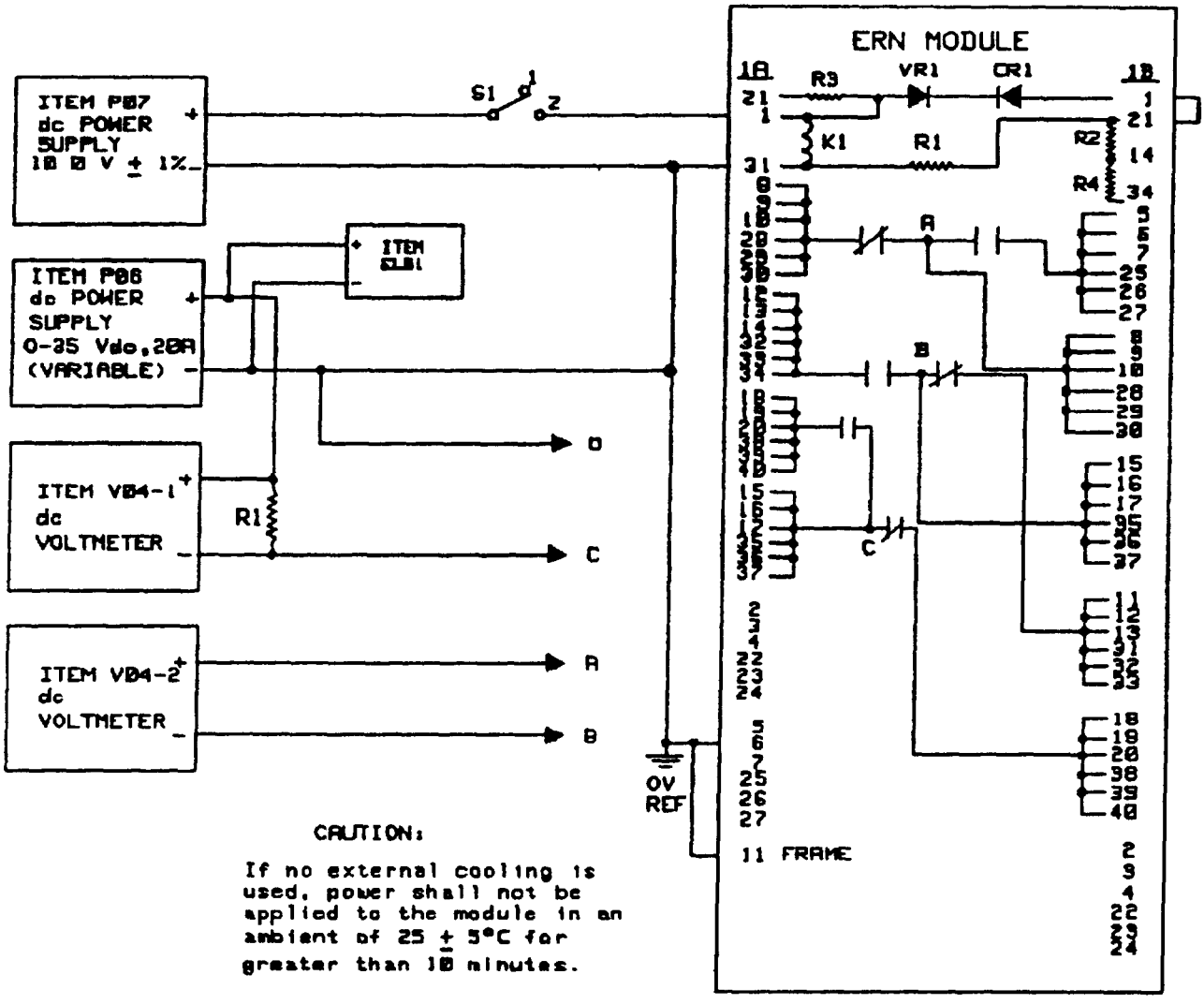


FIGURE 7. Contact voltage drop test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

NOTES:

1. S1 = SPDT switch.

2. R1 = 1.4 Ohm ± 5 percent, 1,000 W, non-inductive wire wound.

3. Total wire resistance: 200 mOhm maximum between item P06 (0-35 V, 20A) and module.

4. Measure voltage drop (item V04-2) on module pins within 0.5 inch (12.7 mm) of each module pin under test.

5. Since the voltmeters are not used simultaneously, only one item V04 is required.

6. For acceptance testing, any contact current between 2 and 20 amps may be used.

7. For contact currents greater than 3 amps, the 5 pins for each contact not connected to points A or B shall be wired together externally.

8. Test instructions:

    (a) Set item EL01 (figure 7) as follows:

        Range (voltage) 60 volts.
        Range (current) 18 amps.
        Set "mode" to "0-50 V" position.
        Set coarse and fine load adj. knobs fully CCW.

    (b) Short points C and D.

    (c) Adjust item P06 for the desired contact current where $I_{contact} = V_{R1} / 1.4$; V_{R1} = voltage on item V04-1 (voltage drop across R1).

    (d) Open points C and D.

    (e) Slowly adjust item EL01 "load adj. knob" for a 0.5 to 2.0 ampere reading on item EL01.

9. Test instructions for test conditions H1-H3:

    (a) Set switch S1 to position 2.

FIGURE 7. Contact voltage drop test circuit for table IX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Connect points A, B, C, and D to module pins as specified in applicable test condition in table IX.

(c) De-energize relay (set switch S1 in position 1).

(d) Energize and de-energize relay twice using switch S1.

(e) Reading on item V04-2, with switch S1 in position 1, shall meet the limits specified in table IX.

CAUTION: Set switch S1 to position 2 before disconnecting points C and D.

10. Test instructions for test conditions H4-H6:

(a) Set switch S1 to position 1.

(b) Connect points A, B, C, and D to module pins as specified in applicable test condition in table IX.

(c) Energize relay (set switch S1 in position 2).

(d) De-energize and energize the relay twice using switch S1.

(e) Reading on item V04-2, with switch S1 in position 2, shall meet the limits specified in table IX.

CAUTION: Set switch S1 in position 1 before disconnecting points C and D.

FIGURE 7. Contact voltage drop test circuit for table IX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE X. Pickup voltage test.

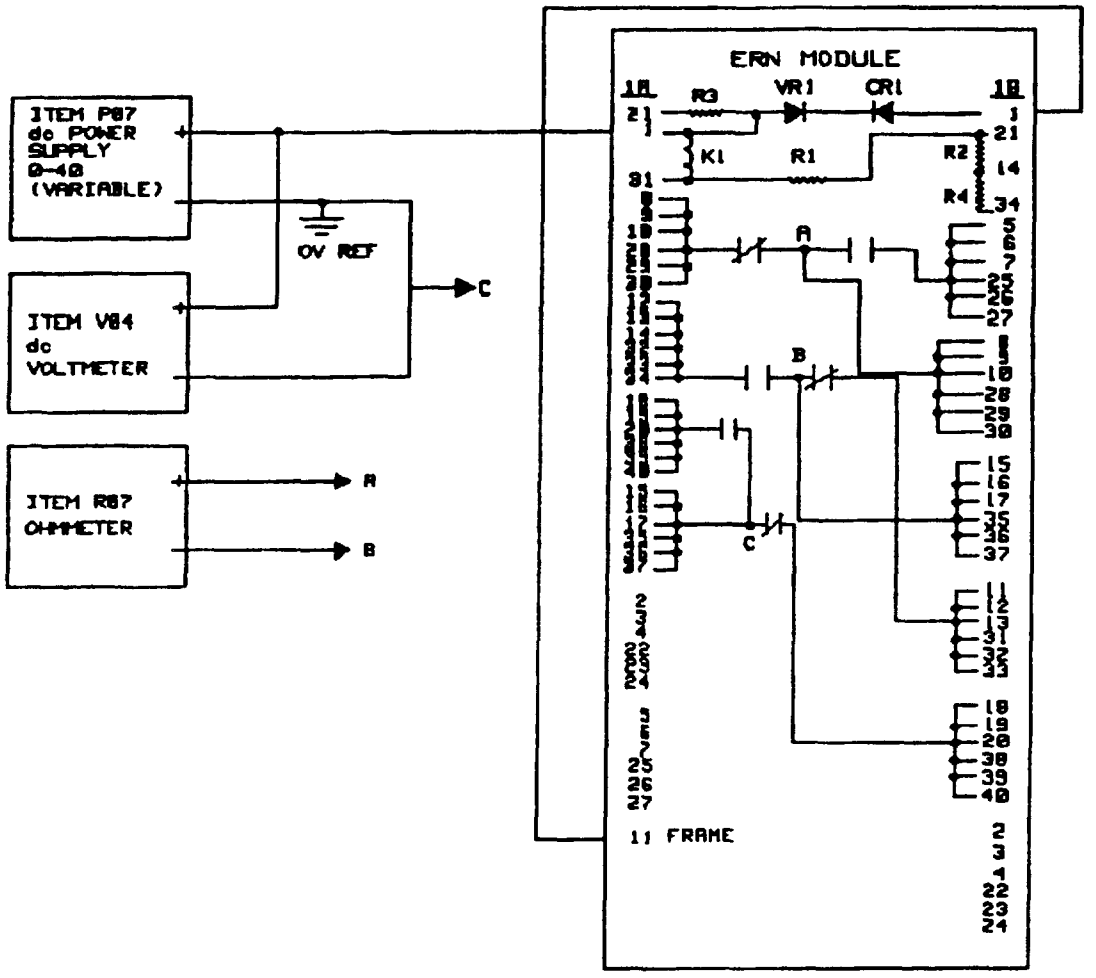
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Common Pins	Common Pins	Initial limits 25 ± 5°C Ambient Min	Initial limits 25 ± 5°C Ambient Max	Unit	Test figure
J1	Pickup voltage	1B5	1A15	1A31	1B8 and 1A12	1B15 and 1A18	3.0	7.2	V	8
J2	Pickup voltage	1B5	1A15	1B34	1B8 and 1A12	1B15 and 1A18	12	25	V	8
J3	Pickup voltage	1B5	1A15	1B14	1B8 and 1A12	1B15 and 1A18	10.0	22.0	V	8
J4	Pickup voltage	1B5	1A15	1B21	1B8 and 1A12	1B15 and 1A18	9	20.0	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE XI Relay dropout voltage test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Common Pins	Common Pins	Initial limits 25 ± 5°C Ambient Min	Initial limits 25 ± 5°C Ambient Max	Unit	Test figure
K1	Dropout voltage	1A8	1B18	1A31	1B8 and 1B15	1B11 and 1A15	0.5	2.0	V	8
K2	Dropout voltage	1A8	1B18	1B24	1B8 and 1B15	1B11 and 1A18	0.5	11.8	V	8
K3	Dropout voltage	1A8	1B18	1B14	1B8 and 1B15	1B11 and 1A15	0.5	8.0	V	8
K4	Dropout voltage	1A8	1B18	1B21	1B8 and 1B15	1B11 and 1A15	0.5	8.0	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.



NOTES:

1. Pickup voltage test: Slowly increase item P07 voltage from 0 volts, but not to exceed specified max pickup voltage +3VDC for T.C. J1-J4 until continuity is observed on item R07. The voltage read on item V04 shall satisfy the requirements in table X.

2. Dropout voltage test: Adjust item P07 voltage to max pickup voltage +3V for T.C. K1-K4. Slowly decrease item P07 voltage until continuity is observed on item R07. The voltage read on item V04 shall satisfy the requirements in table XI.

FIGURE 8. Relay pickup and dropout voltage test circuit for tables X and XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE XII. Relay operate time (including contact bounce) test.

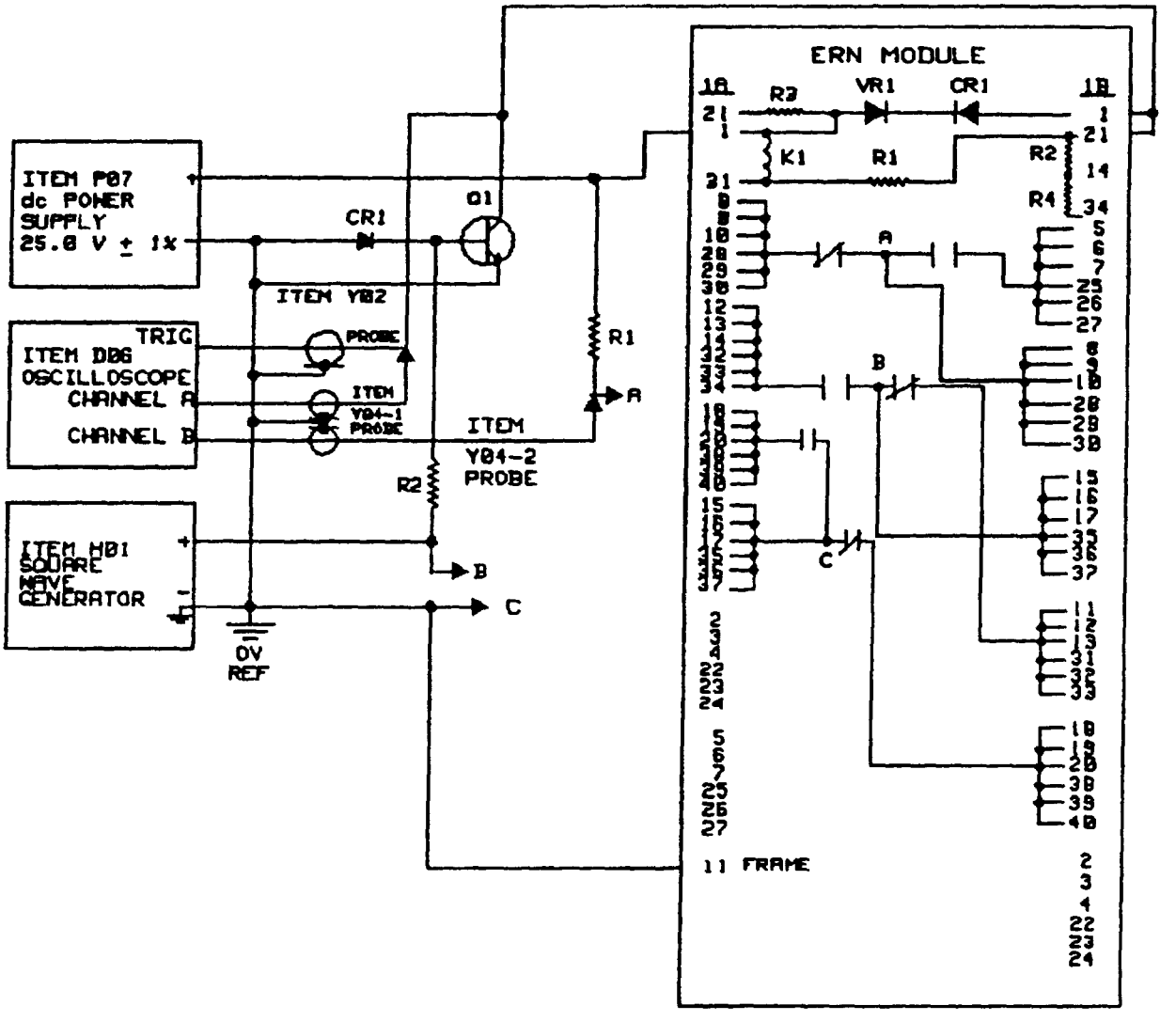
Test condition	Parameter	Connect		Initial limits 25 ± 5°C Ambient		Unit	Test figure
		A to pin	C to pin	Min	Max		
L1	Operate time (ts) A(NO)	1B8	1B5	1.0	20.0	ms	9,10
L2	Operate time (ts) B(NO)	1B15	1B12	1.0	20.0	ms	9,10
L3	Operate time (ts) C(NO)	1A15	1A18	1.0	20.0	ms	9,10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE XIII. Relay release time (including contact bounce) test.

Test condition	Parameter	Connect		Initial limits 23 ± 5°C Ambient		Unit	Test figure
		A to pin	C to pin	Min	Max		
M1	Release time (td) A(NC)	1B9	1A8	2.0	15.0	ms	9,11
M2	Release time (td) B(NC)	1B15	1A11	2.0	15.0	ms	9,11
M3	Release time (td) C(NC)	1A15	1A10	2.0	15.0	ms	9,11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

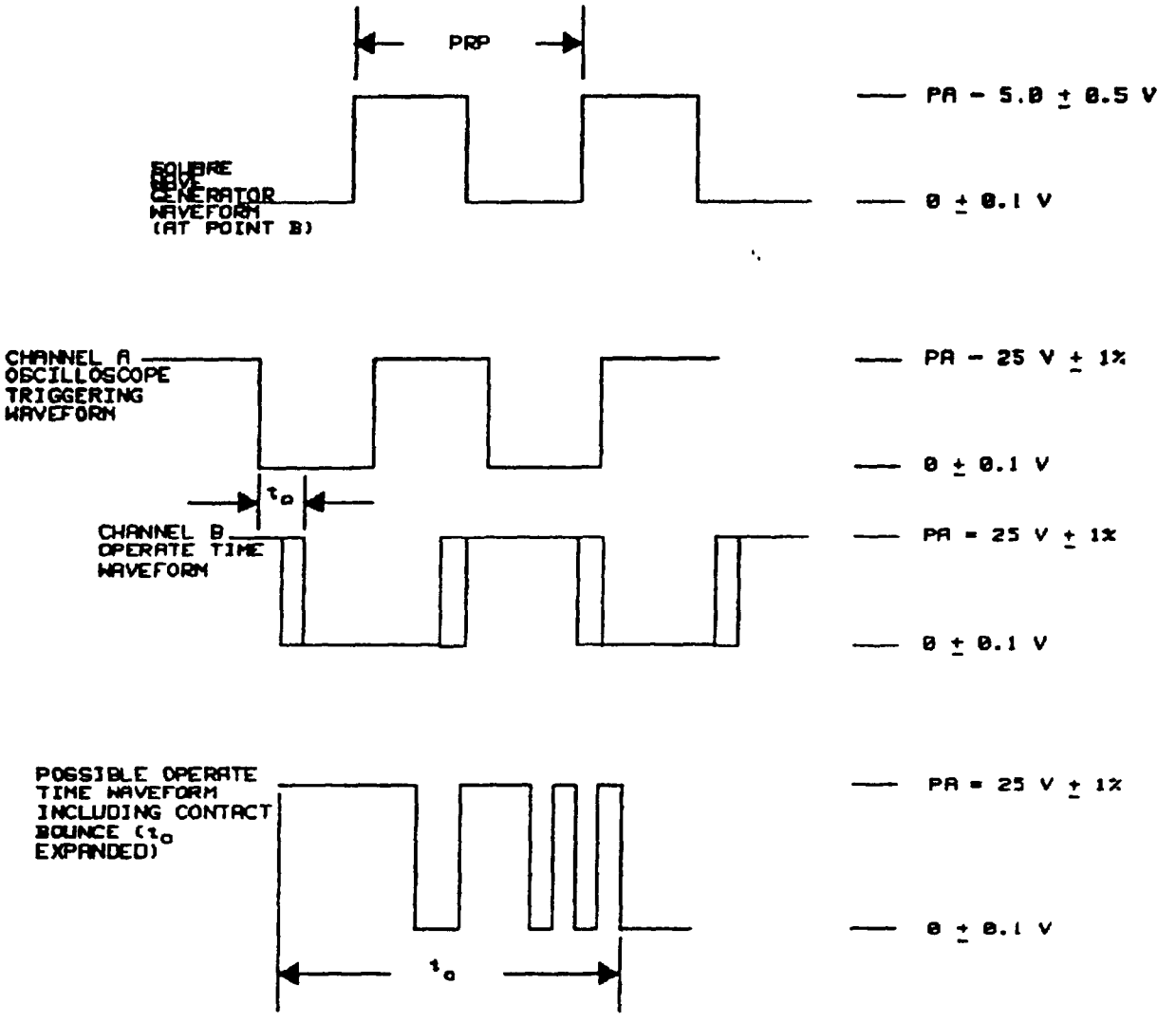


NOTES:

1. R1 = 10 KOhm ± 1 percent, 1/4 W.
2. R2 = 490 Ohm ± 5 percent, 1/4 W.
3. Q1 = 2N3057, or equivalent.
4. CR1 = 1N3612, or equivalent.

FIGURE 9. Relay operate and release time test circuit for tables XII and XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

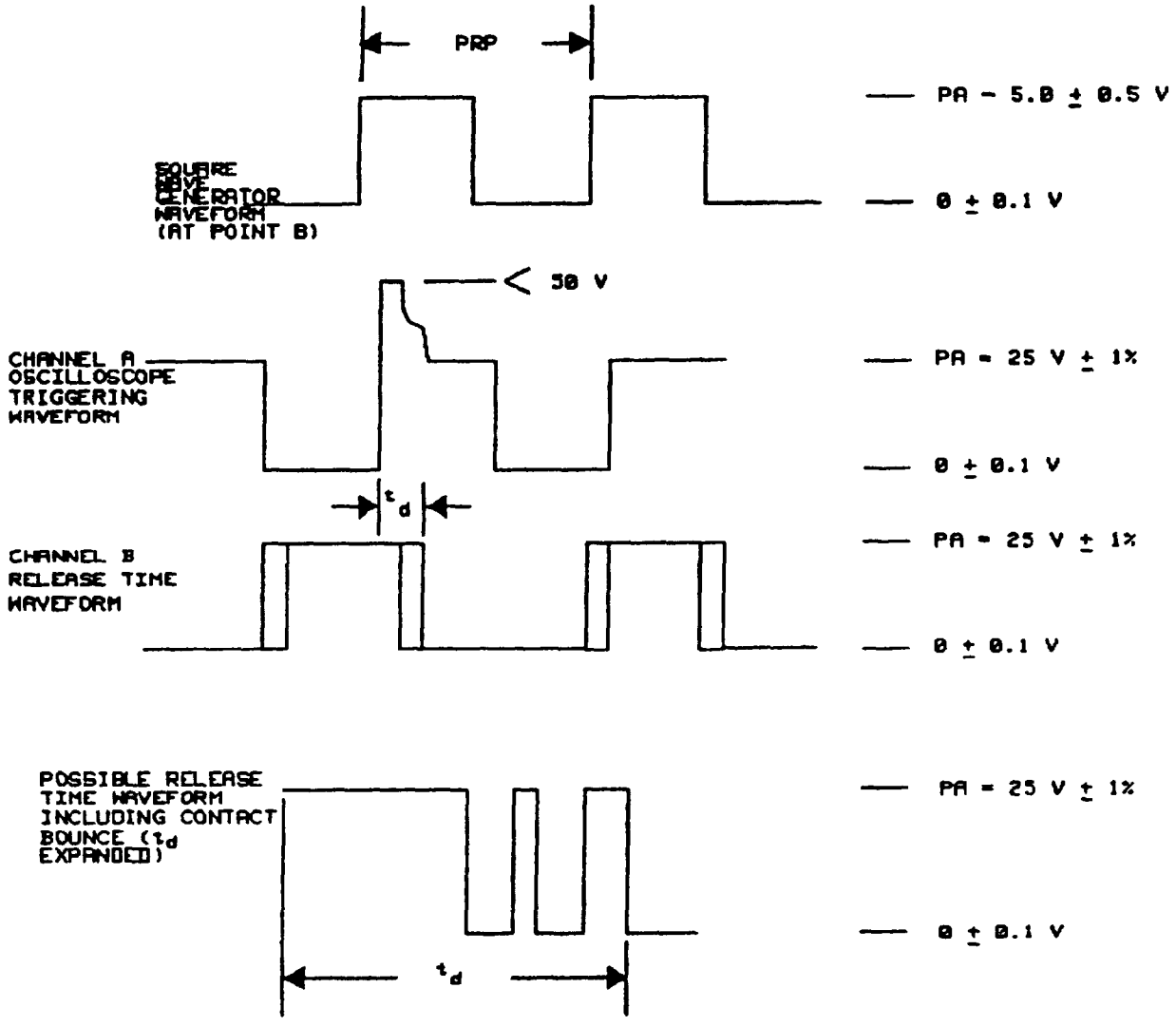


NOTES:

1. PRP = 0.5 ± 0.3 second.
2. All voltages are referenced to 0 V reference.
3. to = Operate time.

FIGURE 10. Relay operate time waveforms for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.



NOTES:

1. PRP = 0.5 ± 0.3 second.
2. All voltages are referenced to 0 V reference.
3. td = Release time.

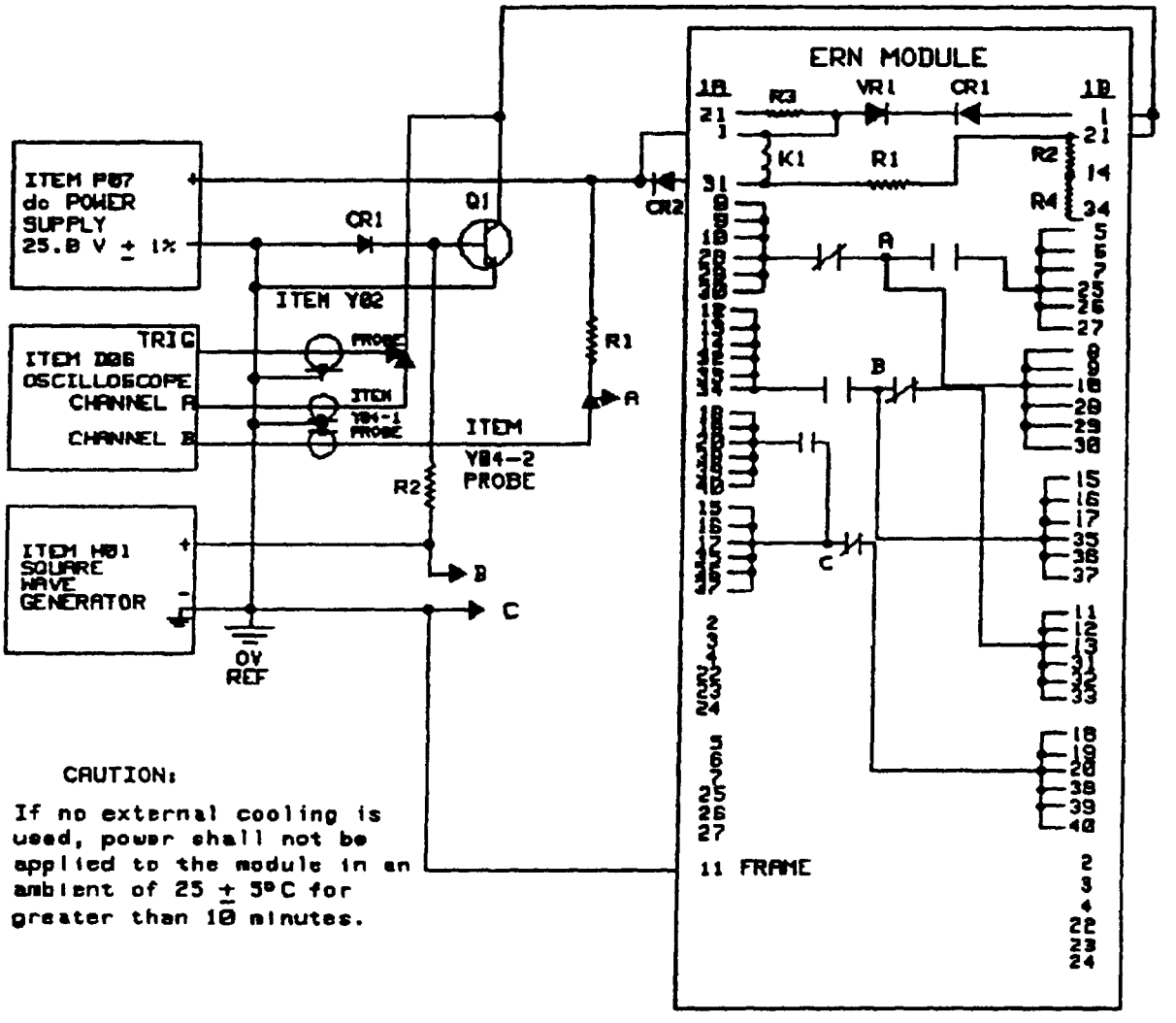
FIGURE 11. Relay release time waveforms for table XIII and XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE XIV. Relay slow release time (including contact bounce) test.

Test condition	Parameter	Connect		Initial limits 25 ± 5°C Ambient		Unit	Test figure
		A to pin	C to pin	Min	Max		
N1	Release time (td) A(NC)	1B8	1R9	8.0	25.0	ms	11,12
N2	Release time (td) B(NC)	1B15	1B11	8.0	25.0	ms	11,12
N3	Release time (td) C(NC)	1A15	1B10	8.0	25.0	ms	11,12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.



NOTES:

1. R1 = 10 KOhm ± 1 percent, 1/4 W.
2. R2 = 490 Ohm ± 5 percent, 1/4 W.
3. Q1 = 2N3057, or equivalent.
4. CR1 = 1N3612, or equivalent.
5. CR2 = 1N3612, or equivalent.

FIGURE 12. Relay release time test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE XV. Contact carry current test.

Test condition	Parameter	Connect		Initial limits 25 ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Contact carry current	1A8	1B8	-	0.360	V	13
P2	Contact carry current	1B11	1B15	-	0.580	V	13
P3	Contact carry current	1B18	1A15	-	0.950	V	13
P4	Contact carry current	1B5	1B8	-	0.360	V	13
P5	Contact carry current	1A12	1B15	-	0.360	V	13
P6	Contact carry current	1A18	1A15	-	0.360	V	13

NOTE:

1. If the requirements in table XV are met, the requirements in table I have been complied with

Table I current (in amps) = $\frac{\text{Voltage obtained from table XV}}{0.015}$

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

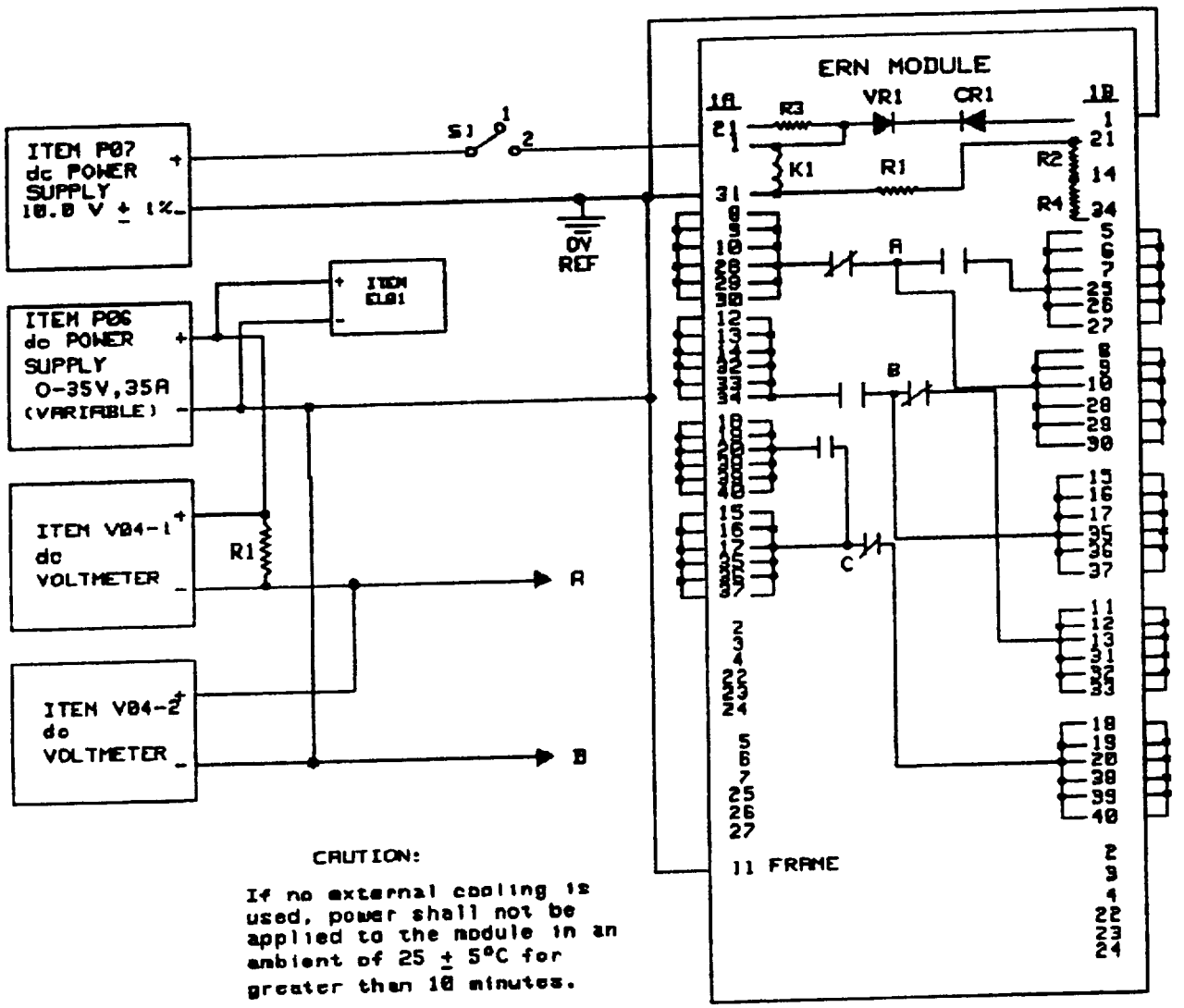


FIGURE 13. Contact carry current test circuit for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

NOTES:

1. S1 = SPDT switch.

2. R1 = 0.3 Ohm ± 5 percent, 400 W.

3. Total wire resistance: 200 mOhm maximum between item P06 (0-35 V, 35 A) and module.

4. Measure voltage drop (item V04-2) on module pins within 0.5 inch (12.7 mm) of each module pin under test.

5. Since the voltmeters are not used simultaneously, only one item V04 is required.

6. Test instructions:

(a) Set item EL01 (figure 13) as follows:

Range (voltage) 60 volts.
Range (current) 18 amps.
Set "mode" to "0 to 50 V" position.
Set course and fine load adj. knobs fully CCW.

(b) Short points A and B.

(c) Adjust item P06 for a reading of 10.5 Vdc ± 1 percent on item V04-1 (voltage drop across R1).

(d) Open points A and B.

(e) Slowly adjust item EL01 load adj. knob for a 0.5 to 2.0 ampere reading on item EL01 ammeter.

7. Test instructions for test conditions P1-P3:

(a) Set switch S1 to position 1.

(b) Connect points A and B to module pins as specified in applicable test condition in table XV.

(c) The reading on V04-2 after 10 minutes time interval for each test condition shall meet the limits specified in table I.

FIGURE 13. Contact carry current test circuit for table XV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

8. Test instructions for test conditions P4-P6:

(a) Set switch S1 to position 2.

(b) Connect points A and B to module pins as specified in applicable test condition in table XV.

(c) The reading on item V04-2 after 10 minutes time interval for each test condition shall meet the limits specified in table I.

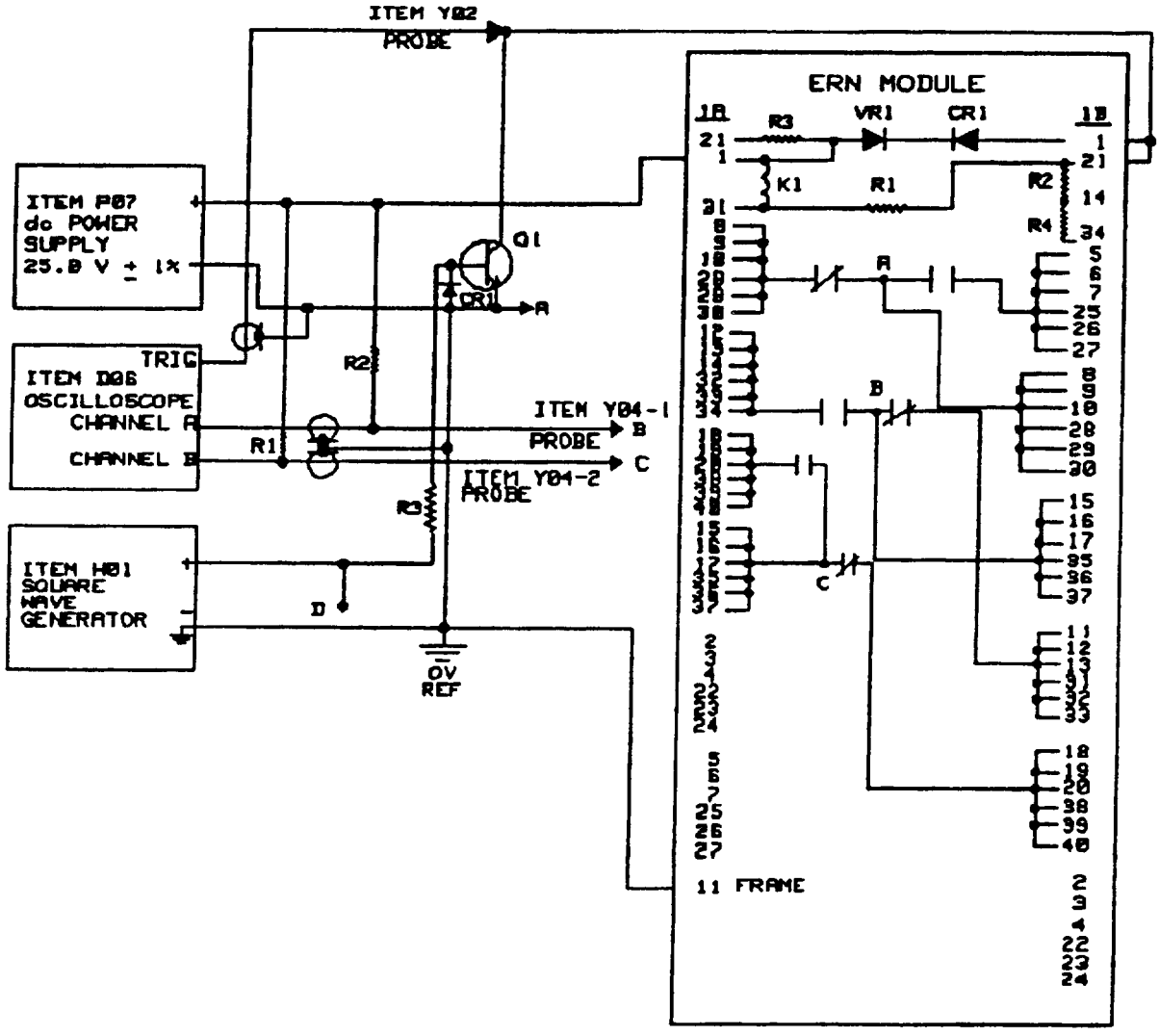
FIGURE 13. Contact carry current test circuit for table XV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE XVI. Relay break before make time test.

Test condition	Parameter	Connect			Initial limits 25 ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
01	A	1A8	1A9	1B5	0.1	5.0	mS	14,15
02	B	1B15	1B11	1A12	0.1	5.0	mS	14,15
03	C	1A15	1B10	1A10	0.1	5.0	mS	14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

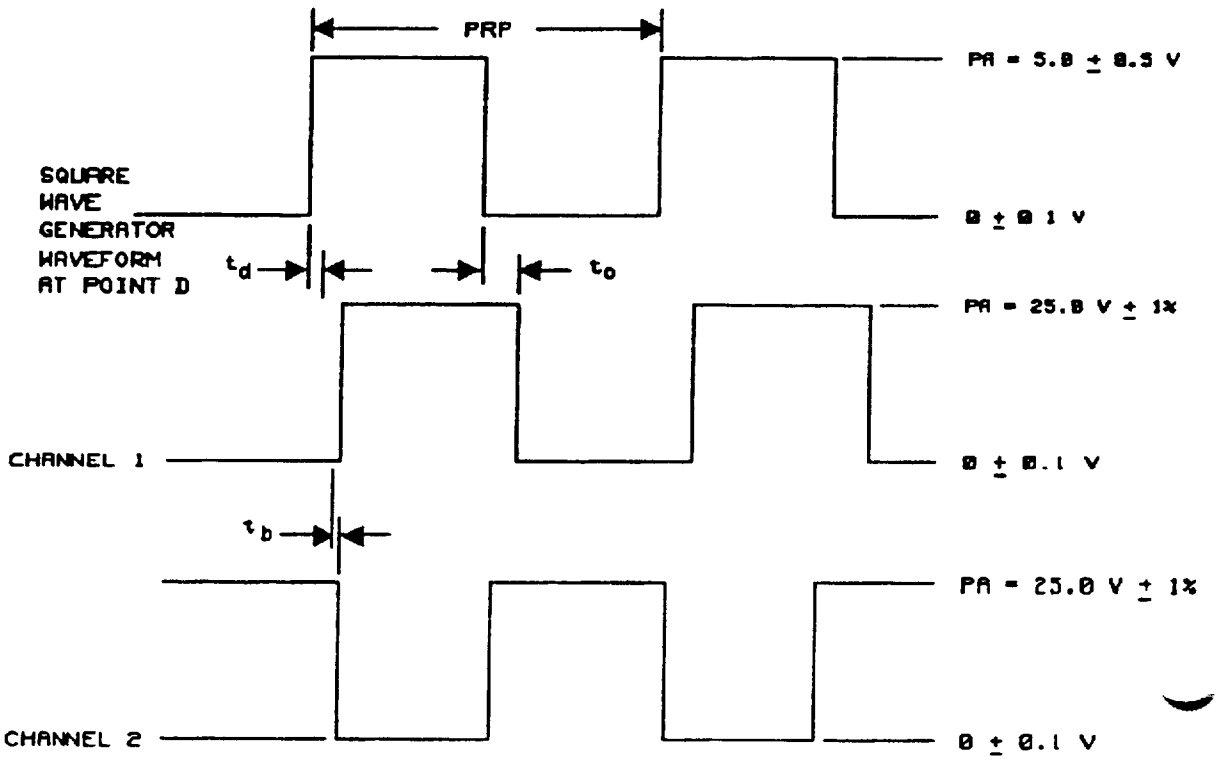


NOTES:

1. R1 = R2 = 10 KOhm ± 1 percent, 1/4 W.
2. R3 = 490 Ohm ± 5 percent, 1/4 W.
3. Q1 = 2N3057, or equivalent.
4. CR1 = 1N3612, or equivalent.

FIGURE 14. Relay break before make time test circuit for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.



NOTES:

1. PRP = 0.5 ± 0.3 second.
2. All voltages are referenced to 0 V reference.
3. tb = Break before make time.
4. to = Operate time (approximately 20 mS, for reference only).
5. td = Release time (approximately 15 mS, for reference only).

FIGURE 15. Relay break before make waveforms for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

TABLE XVII. Sample size (overload and life tests).

Test	Qualification
Over current	Performed following the CCT-TCCT test of MIL-M-28787.
Overload	Performed following the fungus test of MIL-M-28787.
Life test 1	As specified in MIL-M-28787.
Life test 2	Performed following the corrosion resistance test of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

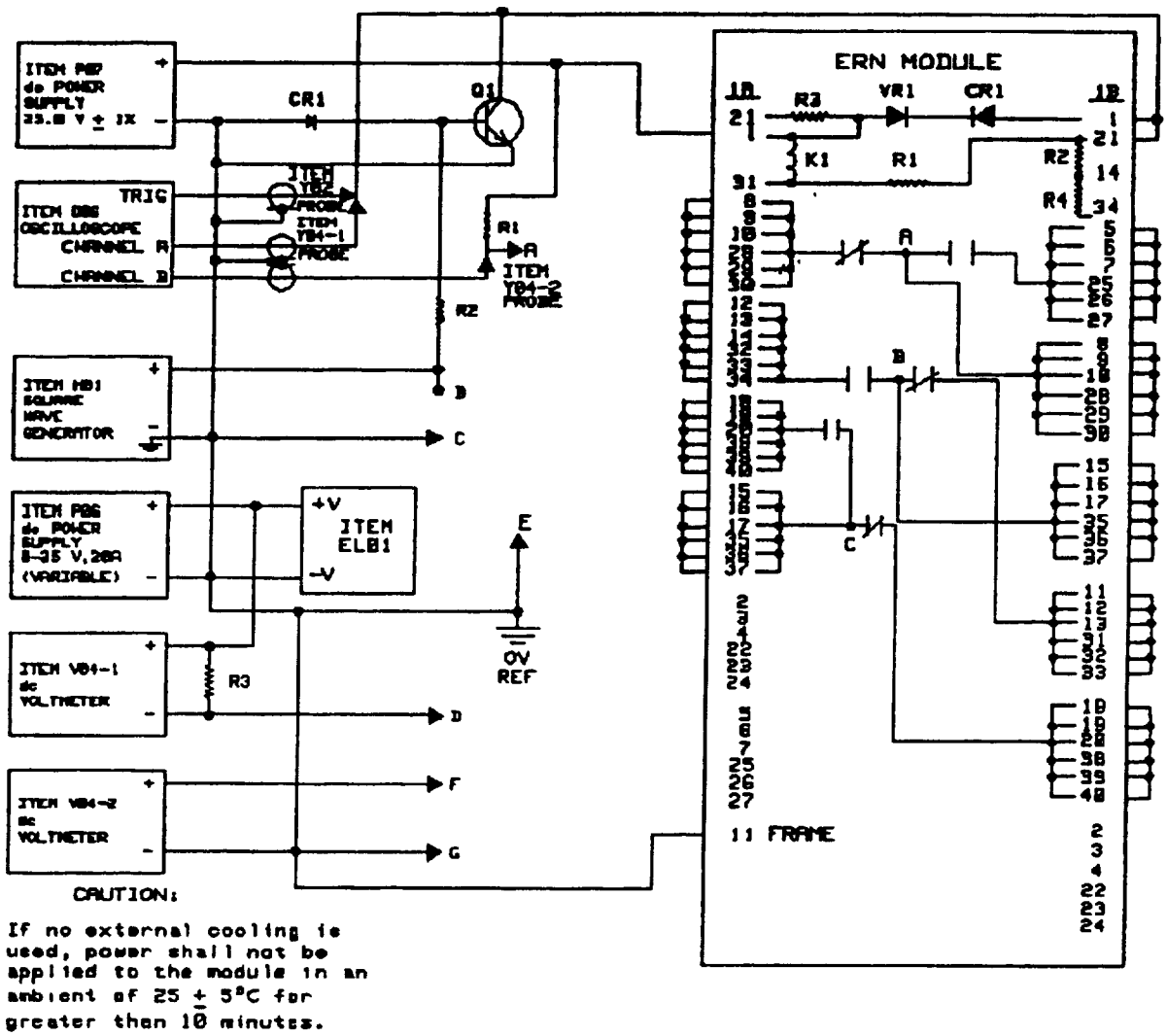


FIGURE 16. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

NOTES:

1. R1 = 10 KOhm ± 1 percent, 1/4 W.

2. R2 = 490 Ohm ± 5 percent, 1/4 W.

3. R3 = 1.6 Ohm ± 5 percent, 1,000 W, noninductive wire wound.

4. Q1 = 2N3057, or equivalent.

5. CR1 = 1N3612, or equivalent.

6. Total wire resistance: 200 mOhm maximum between item P06 and module.

7. Electronic load item EL01 setup.

   (a) Range (voltage) 60 V.
   Range (current) 18 A.
   "Mode" 0-50 V position.
   DC load adjust - fully CCW.

   (b) Using position F(+) and G(-) adjust item P06 for 33.0 ± 0.5 Vdc.

   (c) Disconnect points F and G.

   (d) Slowly adjust item EL01 "DC load adjust" potentiometer for a 0.5 to 2.0 ampere reading on item EL01 ammeter.

8. Adjust item H01 for an amplitude of 5.0 ± 0.5 V.

9. Measure the contact voltage drop on the following groups of module pins by connecting points D(+) and E(-) as needed and measuring the points F(+) and G(-).

Group	(+)	(-)
1	1A8,1A9,1A10,1A28,1A29,1A30	1B5,1B6,1B7,1B25,1B26,1B27
2	1A8,1A9,1A10,1A28,1A29,1A30	1B8,1B9,1B10,1B28,1B29,1B30
3	1A12,1A13,1A14,1A32,1A33,1A34	1B15,1B16,1B17,1B35,1B36,1B37
4	1A12,1A13,1A14,1A32,1A33,1A34	1B11,1B12,1B13,1B31,1B32,1B33
5	1A18,1A19,1A20,1A38,1A39,1A40	1B18,1B19,1B20,1B38,1B39,1B40
6	1A15,1A16,1A17,1A35,1A36,1A37	1B18,1B19,1B20,1B38,1B39,1B40

10. Since the voltmeters are not used simultaneously, only one item V04 is required.

FIGURE 16. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

11. For operating temperature tests, R3 may be a 16.0 Ohm ± 5 percent, 100 W resistor. Each contact shall carry 2 amps during the operating temperature tests.

12. The closing of contacts should be monitored continually during life tests.

FIGURE 16. Operating temperature and life test circuit - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-28787/86B	27 November 1991

3. DOCUMENT TITLE
Modules, Standard Electronic, Relay, Power, Key Code ERN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

NAME	b. TELEPHONE (Include Area Code)	
Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A)	(1) Commercial	(2) AUTOVON
PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 812-854-1528	8-482-1528
c. ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 — Previous editions are obsolete — 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-29T18:01Z
Check the source to verify that the document is the current revision before use.